

Object-Centric VQ Encoder Architecture

① Input

Observation ($B \times 7 \times 7 \times 3$)

channels: type $\in [0,10]$, color $\in [0,5]$, state $\in [0,2]$

② Embed

Cell Embedding $\text{cat}(\text{type_embed}, \text{color_embed}, \text{state_embed}) \rightarrow 64\text{-dim}$

dim: 22 + 21 + 21

③ Tokenise

[CLS] token + 49 spatial tokens + positional embeddings $\rightarrow 50 \times 64$

`nn.Embedding(50, 64)`

④ Transformer

Transformer Encoder

L=2 layers · H=4 heads · d_ff=256 · dropout=0.1

output: $h[0]$ = CLS token · $h[1..49]$ = per-cell spatial tokens

Agent head

Key head

Door head

⑤ Readout

CLS output $h[0]$

Key cell readout $h[i_{\text{key}}]$

Door cell readout $h[i_{\text{door}}]$

global state token
64-dim

scan obs for type=5
 $\rightarrow h[i_{\text{key}}]$ or key_absent_embed

scan obs for type=4
 $\rightarrow h[i_{\text{door}}]$ or door_absent_embed

⑥ VQ heads

VQ_agent
(K = 64)

VQ_key
(K = 16)

VQ_door
(K = 8)

EMA + dead-code restart

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⑦ Discrete codes

$z_q_agent \cdot \text{idx_agent}$

$z_q_key \cdot \text{idx_key}$

$z_q_door \cdot \text{idx_door}$

64-dim · $\text{idx} \in \{0 \dots 63\}$

64-dim · $\text{idx} \in \{0 \dots 15\}$

64-dim · $\text{idx} \in \{0 \dots 7\}$

⑧ Concat

concat($z_q_agent, z_q_key, z_q_door$) $\rightarrow$ 192-dim policy input

⑨ Decoder

key_aux_head
 $\rightarrow$ CE on key cell

Decoder $\text{Linear}(192 \rightarrow 256) \rightarrow \text{ReLU} \rightarrow \text{Linear}(256 \rightarrow 980) \rightarrow 7 \times 7 \times 20$ logits

door_aux_head
 $\rightarrow$ CE on door cell

L_{recon} (CE, $7 \times 7 \times 3$)

$L_{\text{total}} = L_{\text{recon}} + \lambda(L_{\text{key_aux}} + L_{\text{door_aux}}) + L_{\text{commit}}$ (training)

training only

